

Database Management System CO1-AT2 Concept Map.

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DBMS SOFTWARE ARCHITECTURE

Flow: User Request to Data Retrieval.



1. User / Applications.
User sends request
(e.g: SQL Query)

Query PROCESSOR

2. PARSER



Checks query
syntax and
Semantics.

3. OPTIMIZER



chooses the best
execution plan.

4. EXECUTION
ENGINE

Executes the
Plan.

RESULT



Results returned
to users

STORAGE MANAGER

5. TRANSACTION
MANAGER



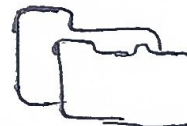
Ensures ACID
Properties

6. AUTHORIZATION &
INTEGRITY MANAGER



Checks permissions
and integrity
constraints.

7. File Manager



Manages files
indexes and
access paths.



DISK STORAGE

Actual data stored on disk
(Files, Pages, indexes, logs).

Key benefits

- * Efficient Processing
- * Data Security & Integrity
- * Reliable Transactions
- * Data Abstraction & Independence.

Three - Level ANSI / SPARC ARCHITECTURE with Data Independence.

Goal

Separate user views and logical structure from physical storage.

EXTERNAL LEVEL
(View Level)
user-specific views
of the database

CONCEPTUAL LEVEL
(Logical Level)
Global logical structure
of the entire database.

INTERNAL LEVEL
(Physical Level)
How data is physically
stored

Key Points

- * Multiple views
- * Shows only requires data
- * Hides the rest

Key Points.

- * Describe entities attributes and relationships.
- * Independent of physical storage

Key Points.

- * Describe storage structures.
- * File organization
- * Indexes, access paths.

LOGICAL DATA INDEPENDENCE

Changes in Conceptual level do not effect external views.

PHYSICAL DATA INDEPENDENCE

Changes in physical level do not effect Conceptual level.